PROBLEM SET 10 DUE 6.7

1. Let A be an integral domain with field of fractions K. Show that

$$A = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} A_{\mathfrak{p}} = \bigcap_{\mathfrak{m} \in \operatorname{Specm} A} A_{\mathfrak{m}}$$

where A, as well as the $A_{\mathfrak{p}}$ for $\mathfrak{p} \in \operatorname{Spec} A$, are viewed as subrings of K.

Hint: for any $\frac{b}{a} \in K$ with $a, b \in A$ and $a \neq 0$, consider the ideal of denominators

$$\mathfrak{a} \coloneqq ((a) : (b)) = \big\{ x \in A \,\big|\, bx \in (a) \big\}.$$

2. Let A be a ring, $S \subseteq A$ multiplicative, M an A-module. Prove the following statements:

(1) If M is a finitely generated A-module, then $S^{-1}M$ is a finitely generated $S^{-1}A$-module.
(2) If M is a finitely presented A-module, then $S^{-1}M$ is a finitely presented $S^{-1}A$-module.
(3) If M is a Noetherian A-module, then $S^{-1}M$ is a Noetherian $S^{-1}A$-module.

3. Let $A = \mathbb{Z}$ and $M = \bigoplus_{p \text{ prime}} \mathbb{Z}/p\mathbb{Z}$. Show that $M_{\mathfrak{p}}$ is a finitely generated $\mathbb{Z}_{\mathfrak{p}}$-module for every $\mathfrak{p} \in \operatorname{Spec} \mathbb{Z}$, but M is *not* a finitely generated $\mathbb{Z}$-module. [1]

4. Let $A = \mathbb{F}_2^{\mathbb{N}}$ be the infinite direct product of $\mathbb{F}_2$ over $\mathbb{N}$. Show that $A_{\mathfrak{p}}$ is a Noetherian ring for every $\mathfrak{p} \in \operatorname{Spec} A$, [2] but A is *not* a Noetherian ring.

Hint: A ring is **Boolean** *if $x^2 = x$ for all $x \in A$. Show that in a Boolean ring A, every prime ideal $\mathfrak{p}$ is maximal and $A_{\mathfrak{p}} \cong A/\mathfrak{p} \cong \mathbb{F}_2$.*

5. Let A be a ring, $S \subseteq A$ multiplicative, $\mathfrak{p} \in \operatorname{Spec} A$ such that $\mathfrak{p} \cap S = \varnothing$. Prove the following statements:

(1) $(S^{-1}A)_{S^{-1}\mathfrak{p}} \cong A_{\mathfrak{p}}$ as rings.
(2) If M is an A-module, then $(S^{-1}M)_{S^{-1}\mathfrak{p}} \cong M_{\mathfrak{p}}$ as $A_{\mathfrak{p}}$-modules. [3]

6. Let A be a ring and let $f_1, \ldots, f_n$ be elements of A that generate the unit ideal, i.e.

$$(f_1, \ldots, f_n) = (1). \tag{$\heartsuit$}$$

Let M be an A-module. Prove the following statements:

(1) ($\heartsuit$) holds if and only if $\operatorname{Spec} A = D(f_1) \cup \ldots \cup D(f_n)$.
(2) $M = 0 \iff M_{f_i} = 0$ for all i.
(3) M is a finitely generated A-module $\iff M_{f_i}$ is a finitely generated A_{f_i}-module for all i.
(4) M is a finitely presented A-module $\iff M_{f_i}$ is a finitely presented A_{f_i}-module for all i.
(5) M is a Noetherian A-module $\iff M_{f_i}$ is a Noetherian A_{f_i}-module for all i.
(6) A is a Noetherian ring $\iff A_{f_i}$ is a Noetherian ring for all i.

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[1] Since $\mathbb{Z}$ and $\mathbb{Z}_{\mathfrak{p}}$ are all Noetherian rings, and for modules over Noetherian rings, f.g., f.p., and Noetherian are equivalent, this example shows that none of f.g., f.p., Noetherian is a local property for modules.

[2] Beware that a prime ideal of $\mathbb{F}_2^{\mathbb{N}}$ need *not* be of the form $\mathbb{F}_2 \times \ldots \times \mathbb{F}_2 \times 0 \times \mathbb{F}_2 \times \mathbb{F}_2 \times \ldots$.

[3] In particular, if $\mathfrak{p} \subseteq \mathfrak{q}$ are prime ideals in A, then $(A_{\mathfrak{q}})_{\mathfrak{p}A_{\mathfrak{q}}} \cong A_{\mathfrak{p}}$ as rings and $(M_{\mathfrak{q}})_{\mathfrak{p}A_{\mathfrak{q}}} \cong M_{\mathfrak{p}}$ as $A_{\mathfrak{p}}$-modules. If $f \notin \mathfrak{p}$, then $(A_f)_{\mathfrak{p}A_f} \cong A_{\mathfrak{p}}$ as rings and $(M_f)_{\mathfrak{p}A_f} \cong M_{\mathfrak{p}}$ as $A_{\mathfrak{p}}$-modules.

7. Let $(A, \mathfrak{m})$ be a local ring and M, N be finitely generated A-modules. Show that if

$$M \otimes_A N = 0$$

then $M = 0$ or $N = 0$.

Hint: Base-change to $k = A/\mathfrak{m}$.

8. Let $(A, \mathfrak{m})$ and $(B, \mathfrak{n})$ be local rings, and let $f : (A, \mathfrak{m}) \to (B, \mathfrak{n})$ be a local homomorphism. [4] Show that if M is a finitely generated A-module such that

$$M_B \coloneqq B \otimes_A M = 0$$

then $M = 0$. [5]

[4] That is, f is a ring homomorphism and $f(\mathfrak{m}) \subseteq \mathfrak{n}$.

[5] When $(B, \mathfrak{n}) = (A/\mathfrak{m}, 0)$, this is Nakayama's lemma.